
Lecture 7: Extensive-Form and Dynamic Games

Author: Daniel Jaramillo Calderon

2026

Contents

1	Sequential Strategic Situations	2
2	Extensive-Form Games	2
2.1	Example: Take-Away Pennies	2
2.2	Game Trees	3
3	Strategies in Extensive Form	3
4	Multi-Stage Games with Observed Actions	3
5	Backward Induction	4
5.1	Two Examples	4
5.2	Stackelberg Competition	5
6	Subgames and Subgame-Perfect Equilibrium	5
7	Imperfect Information	6
7.1	Static Game as Extensive Form	6
8	Summary	6
9	Incomplete Information: Bayesian Nash Equilibrium	6
9.1	Cournot Competition with Private Costs	6
9.2	Bertrand Competition with Private Costs (Discrete Prices)	7

1. Sequential Strategic Situations

Many strategic situations unfold in *sequence*: chess, take-away pennies, Stackelberg competition, market entry. Extensive-form (dynamic) games capture both the timing of moves and the information each player has when she moves.

Two informational regimes will concern us:

- **Perfect information**: every player, when called to move, observes everything that has happened so far.
- **Imperfect information**: some past actions are hidden when later players choose.

2. Extensive-Form Games

An **extensive-form game** Γ consists of:

1. A finite set of players N .
2. A set of actions A .
3. A set of nodes (histories) X with:
 - an initial node x_0 ,
 - every $x \in X$ of the form $x = (a_1, a_2, \dots, a_k)$ with $a_i \in A$,
 - $A(x) \equiv \{a \in A \mid (x, a) \in X\}$ = the set of actions available at x .
4. A probability distribution π governing Nature's moves.
5. A set of end nodes.
6. A move rule: $\iota : X \rightarrow N$ specifying which player moves at each non-terminal x . Let X_i denote i 's decision nodes.
7. An information partition: $I(x)$ is what the mover at x knows about the history.

2.1. Example: Take-Away Pennies

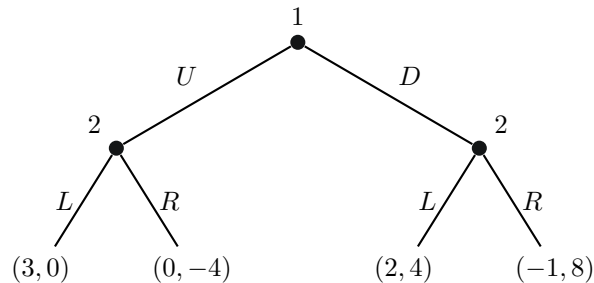
Example

21 pennies on a table. Players alternate removing 1, 2, or 3 pennies; passing is forbidden. Whoever takes the *last* penny loses.

- $N = \{1, 2\}$, $A = \{r_1, r_2, r_3\}$.
- Sample history: $x = (r_1, r_2, r_1, r_3, r_3)$.
- $\pi = 1$ (Nature does not play).
- Move rule alternates between players.
- $I(x) = \{x\}$ — perfect information.

2.2. Game Trees

We draw a game as a tree: black dots are decision nodes (information sets), edges are actions, leaves are end nodes with payoff vectors. When every information set is a singleton, the game has perfect information.



3. Strategies in Extensive Form

In a static game, “strategy” and “action” are interchangeable. In a dynamic game they are not.

A **pure strategy** for player i in extensive-form game Γ is a function

$$s_i : I_i \longrightarrow A,$$

specifying what i would do at *every* information set I_i at which she might be called to move, *including those that are not reached on the equilibrium path*.

Why include unreachable nodes? Because the threat of off-path behaviour disciplines on-path behaviour. “Remove one coin if the number remaining is odd, two if even” is a strategy; “Remove two coins” is an action.

Take-away pennies, 21 coins. The losing positions are 1, 5, 9, 13, 17, 21. From the strategy “Whenever possible, leave the opponent at one of those positions; otherwise remove one,” the second player guarantees a win no matter how player 1 plays.

4. Multi-Stage Games with Observed Actions

A natural special class: at each stage, all active players move *simultaneously*, and at the end of the stage everyone observes the action profile.

Notation.

- Stages indexed $k = 0, 1, \dots, K$.
- Actions in stage k : $a^k = (a_1^k, \dots, a_N^k)$.
- History after stage $k - 1$: $h^k = (a^0, a^1, \dots, a^{k-1})$.
- Available actions at history h^k : $A_i(h^k)$.

A strategy is a contingent plan $\{s_i^k\}_{k=0}^K$ specifying an action $s_i^k(h^k) \in A_i(h^k)$ for every reachable history.

Remark. Stages are not time periods; multiple stages can fit within a single calendar period (e.g. Rubinstein–Stahl bargaining).

A strategy profile s^* is a **Nash equilibrium** if for every player i and every strategy s_i :

$$u_i(s_i^*, s_{-i}^*) \geq u_i(s_i, s_{-i}^*).$$

5. Backward Induction

In games of perfect information we can solve from the end of the tree backwards.

For a finite extensive-form game Γ of perfect information, a joint pure strategy s is constructed as follows. Call a node x **penultimate** if all immediately following nodes are terminal. For every penultimate x , let $s_{i(x)}(x)$ be an action that maximizes the mover's payoff at x , with resulting payoff u_x . Replace the subtree below x by the leaf u_x and repeat until every decision node has been assigned an action.

If s is a backward-induction strategy for a finite extensive-form game of perfect information, then s is a Nash equilibrium.

Consequently, every such game has at least one pure-strategy Nash equilibrium.

5.1. Two Examples

Example

Market Entry. An entrant decides In/Out; if In, the incumbent decides Fight/Acquiesce. Payoffs: status quo (0, 2); In + Fight (−1, −1); In + Acquiesce (1, 1).

Backward induction: at the incumbent's node, Acquiesce (1) beats Fight (−1), so the incumbent will acquiesce. Anticipating this, the entrant enters (1 > 0). The backward-induction equilibrium is (Enter, Acquiesce).

Nash $\neq$ Backward Induction. The strategy “Entrant stays Out; incumbent fights if challenged” is also a Nash equilibrium: given the threat, staying Out is best for the entrant; given that Out is played, the incumbent's off-path threat is never tested. But the threat is *non-credible*: if the entrant did enter, the incumbent would strictly prefer to acquiesce. Backward induction rules out such non-credible threats.

Example

A two-by-two-by-two tree. In a game with player 1 moving first (choices L, R), player 2 moving second (choices l, r), player 1 moving again, the backward-induction outcome is the strategy profile such as $((R, R', L''), (r, l'))$, which records the best move at every node where the player must act, including off-path nodes.

5.2. Stackelberg Competition**Example**

Two firms, no production costs, inverse demand $P(Q) = a - Q$ with $Q = q_1 + q_2$.

Stage 2 (firm 2). Observing q_1 , firm 2 maximizes $(a - q_1 - q_2)q_2$:

$$q_2(q_1) = \frac{a - q_1}{2}.$$

Stage 1 (firm 1). Anticipating $q_2(q_1)$, firm 1 maximizes

$$u_1(q_1) = (a - q_1 - q_2(q_1)) q_1 = \frac{1}{2}(a - q_1) q_1.$$

FOC: $q_1^* = \frac{a}{2}$, hence $q_2^* = \frac{a}{4}$ and $Q^* = \frac{3a}{4}$.

Compared with Cournot's $q^C = a/3$, the Stackelberg leader produces more and the follower less — the leader extracts a first-mover advantage.

6. Subgames and Subgame-Perfect Equilibrium

Backward induction generalises to imperfect-information games via the notion of *subgame*.

Given a multi-stage game with observed actions, for every history h^k the game-from-history- h^k , denoted $G(h^k)$, is itself a well-defined extensive-form game. We call it a **subgame** of the original.

If s is a strategy profile in the full game, $s|_{h^k}$ denotes its restriction to histories consistent with h^k .

A strategy profile s^* is a **subgame-perfect equilibrium** if for every history h^k , $s^*|_{h^k}$ is a Nash equilibrium of $G(h^k)$.

Backward induction is the natural algorithm to compute SPE in finite games of perfect information. In games of imperfect information, backward induction is replaced by “solve every subgame as a Nash equilibrium and stitch.”

7. Imperfect Information

When $I(x)$ is not a singleton, the moving player does not perfectly observe earlier moves. In a tree, an information set is depicted by joining nodes with a dashed line.

Example

Used-car market. A seller chooses Repair/Don't Repair, then sets a High/Low price. The buyer learns the *price* but not whether the car was repaired, and then decides Accept/Reject.

- $N = \{S, B\}$.
- $A = \{\text{repair, don't repair, high, low, accept, reject}\}$.
- Sample node: $x = (\text{repair, high})$; the mover at x is the buyer.
- Information set at price "high": $I(x) = \{(\text{repair, high}), (\text{don't repair, high})\}$.

A static game can also be represented in extensive form by drawing two singleton information sets joined by a dashed line.

7.1. Static Game as Extensive Form

A simultaneous-move game can equivalently be drawn as an extensive form by letting one player "move first" but keeping her action hidden inside a non-singleton information set. The two representations are strategically identical — this is why turn order in Cournot does not matter, whereas in Stackelberg (perfect information) it does.

8. Summary

Setting	Solution concept	Tool
Perfect info, finite	Backward induction	Recursive optimisation
Imperfect info, multi-stage	Subgame-perfect equilibrium	NE of every subgame
Static (one-shot)	Nash equilibrium	Best responses

9. Incomplete Information: Bayesian Nash Equilibrium

In many strategic situations agents do not know each other's payoff-relevant characteristics (costs, valuations, types). We solved the formalism for static games of incomplete information in Lecture 6; here we apply it to two canonical models where the computation is explicit.

9.1. Cournot Competition with Private Costs

Example

Setup. Two firms compete a la Cournot. Inverse demand is $P(Q) = a - Q$, $Q = q_1 + q_2$. Firm 1's marginal cost c is common knowledge. Firm 2 has one of two possible types indexed by $j \in \{1, 2\}$ with costs $c_1 < c_2$; the probability that firm 2 is type j is $\Pr(t_j) = p_j$ (with $p_1 + p_2 = 1$). Firm 2 observes its own type; firm 1 only knows the distribution.

Strategies. The Bayesian game has three players: firm 1, firm 2 of type 1, and firm 2 of type 2. A strategy profile is $(q_1, \tilde{q}_{21}, \tilde{q}_{22})$.

Best responses. For each type j of firm 2:

$$\max_{q_{2j} \geq 0} (a - q_1 - q_{2j})q_{2j} - c_j q_{2j} \implies \tilde{q}_{2j}(q_1) = \frac{a - c_j - q_1}{2}.$$

Firm 1 maximises expected profit over firm 2's types:

$$\max_{q_1 \geq 0} (a - q_1 - \mathbb{E}[\tilde{q}_2])q_1 - c q_1, \quad \mathbb{E}[\tilde{q}_2] = p_1 \tilde{q}_{21} + p_2 \tilde{q}_{22}.$$

FOC:

$$a - 2q_1 - \mathbb{E}[\tilde{q}_2] - c = 0 \implies q_1^* = \frac{a - c - \mathbb{E}[\tilde{q}_2^*]}{2}.$$

Equilibrium. Let $\bar{c}_2 \equiv p_1 c_1 + p_2 c_2$ (expected cost of firm 2). Substituting $\tilde{q}_{2j}^* = (a - c_j - q_1^*)/2$ and solving:

$$q_1^* = \frac{a - 2c + \bar{c}_2}{3}, \quad \tilde{q}_{2j}^* = \frac{a - 2c_j + c}{3} + \frac{c_j - \bar{c}_2}{3}.$$

The triple $(q_1^*, \tilde{q}_{21}^*, \tilde{q}_{22}^*)$ is the unique Bayesian–Nash equilibrium.

9.2. Bertrand Competition with Private Costs (Discrete Prices)

Example

Setup. Firm 1's marginal cost is 0 (common knowledge). Firm 2 has cost $c \in \{c_L = 1, c_H = 4\}$, each equally likely. Market demand is $Q(p) = 8 - p$. Prices are restricted to $\{1, 4, 6\}$. Firm i 's profit when it sets p_i and its rival sets p_j :

$$\pi_i = \begin{cases} Q(p_i)(p_i - c_i), & p_i < p_j, \\ \frac{1}{2}Q(p_i)(p_i - c_i), & p_i = p_j \text{ and } p_i \geq c_i, \\ 0, & p_i > p_j \text{ or } p_i < c_i. \end{cases}$$

Strategic form. Treat the two cost types of firm 2 as separate players — “Firm 2_L ” and “Firm 2_H ”. The game then has three players: Firm 1, Firm 2_L , and Firm 2_H . Firm 1's payoff is the expectation over firm 2's type weighted equally at $\frac{1}{2}$.

Finding the BNE. One can verify by best-response analysis on the resulting matrix game that a Bayesian–Nash equilibrium has Firm 1 set $p_1 = 4$, Firm 2_L set $p_{2L} = 4$, and Firm 2_H set $p_{2H} = 6$.

Private information breaks the Bertrand paradox. With complete information both types would undercut to marginal cost. With private costs, firm 1 cannot perfectly anticipate firm 2's type, and the equilibrium supports positive markups.

Reference: Fudenberg & Tirole, Game Theory; Jehle & Reny, Advanced Microeconomic Theory, Chapter 7.